AI-Powered Connect 4 Proposal

1. Basic Information:

- **Faculty:** Faculty of Computers and Information Mansoura University
- **Department :** Computer Science
- **Course :** [CS324P] Artificial Intelligence - 1
- **Academic Year:** Third Year (CS)
- **Instructor** : Sara El-Metwally
- **Students** : Alia Harb , Shahd Ayman

2. Problem Definition:

Connect 4 is a zero-sum, two-player game.

The challenge is to design an **Intelligent Agent** capable of making strategic decisions to outplay a human opponent. This requires analyzing a 6*7 grid to predict outcomes several moves ahead

The objective is to design an AI agent that can consistently outperform a human player using strategic decision-making.

3. Agent Design (PEAS):

- **Performance:** Achieving victory (4 connected pieces), blocking opponent wins and optimizing computational efficiency .
- **Environment:** A 6*7 grid (board matrix) with a competitive human opponent.
- **Actuators:** Dropping a token into one of the 7 columns.
- **Sensors:** Detecting the current state of the board (identifying empty slots and existing tokens).

4. Environment Properties (ODESDA):

- **Observable: Fully Observable** – The agent has complete access to the board state at any time.
- **Deterministic: Deterministic** – Each move leads to a specific, predictable next state.
- **Episodic: Sequential** – Current moves affect future game possibilities.
- **Static: Static** – The environment does not change while the agent is "thinking".
- **Discrete: Discrete** – There are a limited number of distinct percepts and actions.
- **Agent: Competitive Multi-Agent Environment** – Two agents competing for conflicting goals.

5. Agent Type:

- **Utility-based Agent:** The agent uses an evaluation function (Heuristic) to choose actions that maximize its expected "happiness" (winning score) in future states. This is achieved through adversarial search using the Minimax algorithm.

6. Problem Formulation:

To comply with the formal problem-solving structure:

1. **Initial State:** An empty 6*7 grid represented by a zero-matrix.
2. **Actions:** Selecting any column C in {0, … ,6} that is not yet full.
3. **Transition Model (Successor):** The resulting board state after placing a piece in the selected column.
4. **Goal Test:** A condition where 4 pieces of the same color are connected horizontally, vertically, or diagonally.
5. **Path Cost:** Represented implicitly through a utility function rather than a traditional cost function, where higher values correspond to more favorable states..

7. Methodology & Algorithms:

- **Naive Solution:** A basic GUI with **human vs. human logic.**
- **Refining to AI:** Implementing **Minimax Algorithm** for optimal decision-making.
- **Optimization:** Using **Alpha-Beta Pruning** to eliminate unnecessary branches and deepen the search.
- **Heuristics:** A custom scoring function **that prioritizes** the center column and evaluates 2-piece, 3-piece, and 4-piece sequences.

2. Implementation of Problem Formulation:

```
class Connect4Problem:

    def __init__(self):
        self.initial_state = create_board()
        # Empty 6x7 grid

    def actions(self, state):
        return get_valid_locations(state)
        # Available columns

    def result(self, state, action, piece):
        # Transition Model
        return drop_piece_on_copy
        (state,action, piece)

    def goal_test(self, state, piece):
        return winning_move(state, piece)
        # Check for 4 in a row
```

3. Conclusion:

The development of this AI-powered Connect 4 project serves as a comprehensive application of **Intelligent Agent** design and **Adversarial Search** strategies.
By transitioning from a naive human-vs-human logic to a sophisticated **Utility-based Agent**, we successfully integrated the **Minimax algorithm** with **Alpha-Beta Pruning** to achieve an optimal balance between decision quality and computational efficiency. This project not only demonstrates our understanding of formal **Problem Formulation** but also showcases the practical implementation of **Heuristic Evaluation** to solve complex, real-time strategic challenges.